

Question number	Answer	Additional guidance	Mark
7(a)	A description that includes the following points: • one glycerol and three fatty acids (1) • (joined together by) ester {bonds / links / groups} (1)	ACCEPT joined by esterification	(2)

Question number	Answer	Additional guidance	Mark
7(b)	An explanation that includes three of the following points: • because large lungs would have a large volume of air {in the alveoli / taken in} (1) • because there would be {more alveoli for gas exchange / large surface area for more gas exchange / larger surface area for gas exchange} (1) • because large heart could pump more blood (1) • therefore supplying cells with sufficient oxygen (1)	ACCEPT more oxygen to enter ACCEPT diffusion of {gases / oxygen / carbon dioxide} ACCEPT pump at higher pressure more warm blood ACCEPT (more warm blood) to maintain body temperature	(3)

Question number	Answer	Additional guidance	Mark
7(c)(i)	• 4.3×10^{-11}	ACCEPT 4.30×10^{-11}	(1)

Question number	Answer	Additional guidance	Mark
7(c)(ii)	• volume of sphere calculated (1) • 1 : 3 / 1 : 2.7 / 1 : 2.67 (1)	2048 2 144.66058450 ACCEPT 1 : 3 / 1 : 2.8 / 1 : 2.79 ecf 1 : 21 / 1 : 21.3 / 1 : 21.33 (if 16 used) 1 : 22 / 1 : 22.3 / 1 : 22.34 (if 16 used) Bald answer of 1 : 3 / 1 : 2.7 / 1 : 2.8 / 1 : 2.67 / 1 : 2.79 = 2 marks 1 : 21 / 1 : 21.3 / 1 : 21.33 / 1 : 22 / 1 : 22.3 / 1 : 22.34 / 2048 / 2 144.66 = 1 mark	(2)

Question number	Answer	Mark
*7(c)(iii)	Indicative content: High concentration of red blood cells: • therefore, they carry high levels of oxygen • to the tissues for (aerobic) respiration • so can meet the oxygen demands of the yaks Small red blood cells: • therefore, the surface area to volume ratio is high • so oxygen can diffuse into the cells faster • although each red blood cell probably carries less oxygen • but this will be compensated by the large number of cells • smaller red blood cells pack closer together than larger ones / more cells in small volume of blood High concentration of haemoglobin: • therefore, high levels of oxygen can be carried • as each red blood cell can carry more oxygen • as each haemoglobin molecule can bind to 4 oxygen molecules Fetal haemoglobin: • which has a higher affinity for oxygen • therefore, the haemoglobin can bind the oxygen even in low partial pressures • and therefore carry more oxygen / have a higher percentage saturation of haemoglobin with oxygen	(6)

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.	Simple links made between characteristics of blood and altitude 1 mark = one comment about one characteristic 2 marks = one comment made about two characteristics
Level 2	3-4	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.	Extended links made characteristics of blood and altitude 3 marks = one comment made about three characteristics 4 marks = extended comments about one characteristic + a comment about one other characteristic
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting and applying relevant biological facts / concepts. Consequences are discussed which supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.	Clear discussions on all aspects of blood and altitude 5 marks = extended comments about two characteristics + a comment on the other two characteristic 6 marks = extended comments about three aspects + a comment on the other characteristic